PACKRISK
SUPPLY RISK ANALYTICS

Supplier Performance

History is the same in every scenario. The red table changes: late rate alone is a poor guide; stock cover decides.

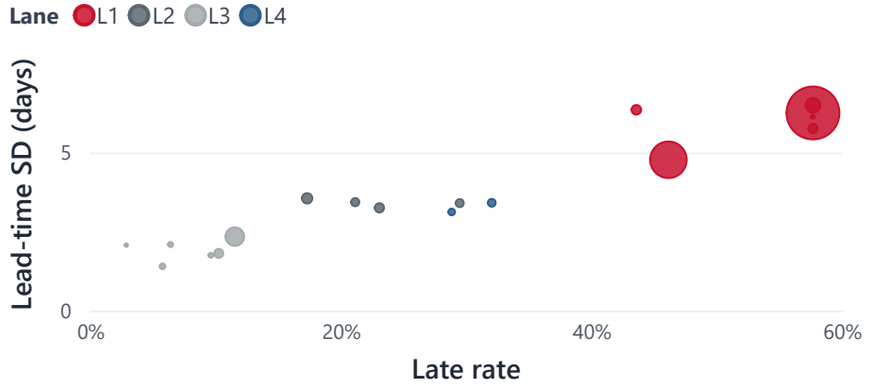
Scenario: Asia ocean-lane port congestion

Average OTIF	Average late rate	Suppliers with history	Single-sourced parts
65.0%	28.8%	18	41

Supplier scorecard (sorted by late rate)

supplier_id	supplies	lane_id	tier2_id	otif_rate	late_rate	sd_lead_time_days	partial_rate
S03	NMC cylindrical cells	L1	T6	42.3%	57.7%	6.25	7.7%
S07	Electronics (CMB, BMS master)	L1	T2	34.6%	57.7%	6.49	15.4%
S15	Mica, aerogel barriers	L1	T5	41.0%	57.7%	5.76	6.4%
S18	Fasteners	L1		35.9%	57.7%	6.13	12.8%
S01	LFP prismatic cells	L1	T6	50.0%	46.2%	4.77	3.9%
S09	Contactors, precharge parts	L1		52.6%	43.6%	6.35	6.4%
S08	Sensors, isolation monitor	L4	T2	60.3%	32.1%	3.41	9.0%
S06	HV cables, harnesses	L2	T1	61.5%	29.5%	3.40	12.8%
S10	Pyro fuse, service disconnect	L4		67.3%	28.9%	3.12	5.8%
S19	Cell frames, holders, plastics	L2	T4	67.3%	23.1%	3.25	11.5%

Late rate vs lead-time variability (size = spend)



Shared dependencies (lanes and tier-2 sources)

Type	ID	Suppliers	Parts	Single-src	Weekly spend	Min cover
Shipping lane	L1	6	13	6	$10,690,200	2.0
Tier-2 source	T6	2	2	0	$9,750,000	4.0
Shipping lane	L3	6	21	19	$1,163,490	2.0
Tier-2 source	T3	2	9	9	$954,000	3.0
Shipping lane	L2	4	11	11	$733,200	2.0
Tier-2 source	T2	2	5	5	$645,600	2.0
Tier-2 source	T4	3	8	6	$427,140	2.0

KEY INSIGHT: parts that stop the line in this scenario (do nothing)

Supplier	Part	Stops line	Late rate	Stock (weeks)
S07	Cell monitoring board	36%	57.7%	2.00
S18	Rivet nut	26%	57.7%	2.00
S18	M8 structural bolt	22%	57.7%	2.00
S07	BMS master controller	15%	57.7%	2.00
S15	Thermal insulation pad	15%	57.7%	2.00
S09	Precharge contactor	12%	43.6%	2.00
S15	Mica fire barrier sheet	2%	57.7%	3.00

Simulated data. Placeholder delay parameters.

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Executive Overview

Battery-pack supply risk. KPIs compare the smart plan (stochastic optimization) with doing nothing, for the selected scenario.

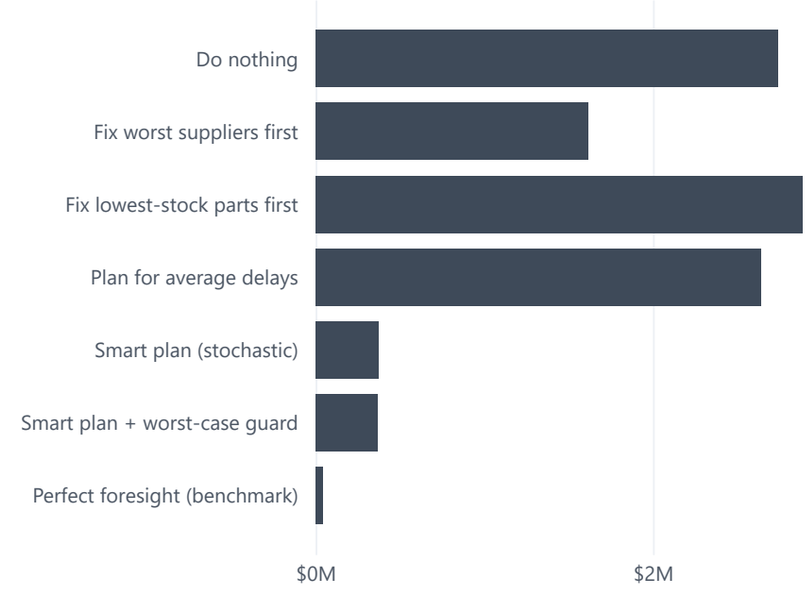
Scenario: Asia ocean-lane port congestion

The smart plan protects 963 packs and saves $2.4M net for $181K of fixes (13.1x return).

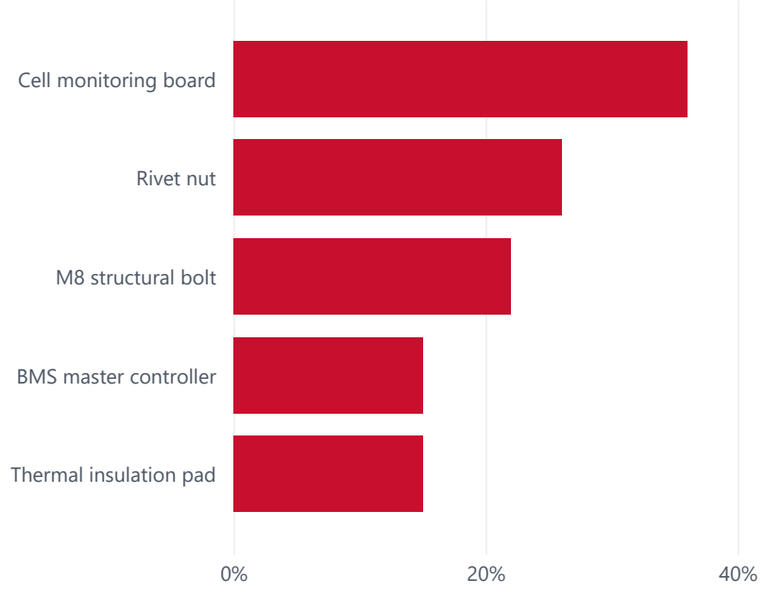
Lost: do nothing	Lost: smart plan	Packs protected	Net savings	Spend on fixes	Saved per $1 spent	Share of best possible
1,037	75	963	$2.4M	$181.2K	13.1x	93.1%

All L1 ocean shipments delayed; hits both cell suppliers, electronics, contactors, fire barriers, fasteners.

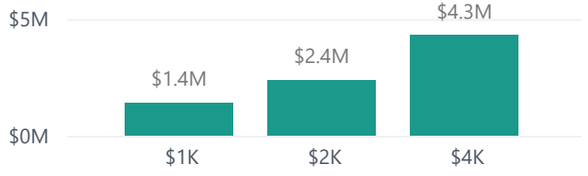
Expected total cost by strategy (lower is better)



Top 5 parts that stop the line (if we do nothing)



Net savings by value of a lost pack



Unseen test scenarios won (H1-H5)

5 of 5

Unseen tests: packs saved per scenario

716

Unseen tests: total net savings

$9.1M

Simulated data. Placeholder delay parameters.

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SUPPLY RISK ANALYTICS

Production Risk

Severe (P90) realization is the default: in the median realization most policies show no shortfall.

Scenario: Asia ocean-lane port congestion

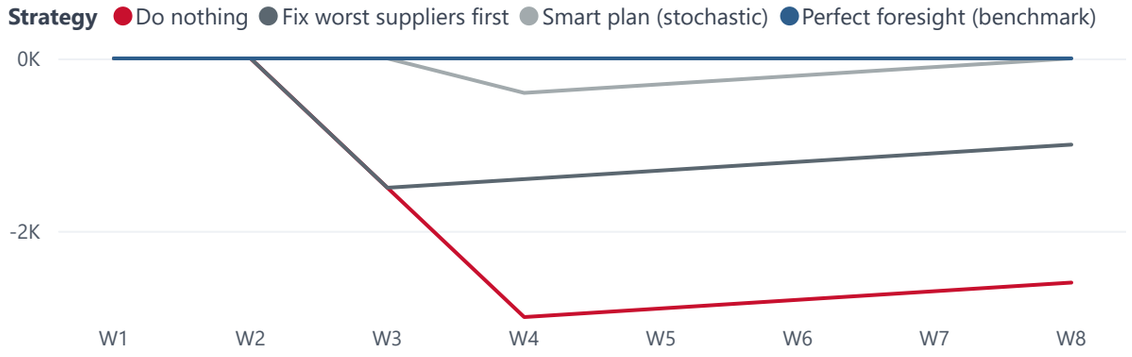
Outcome: Bad case (worst 10%)

Share of plan built (do nothing)
91.4%

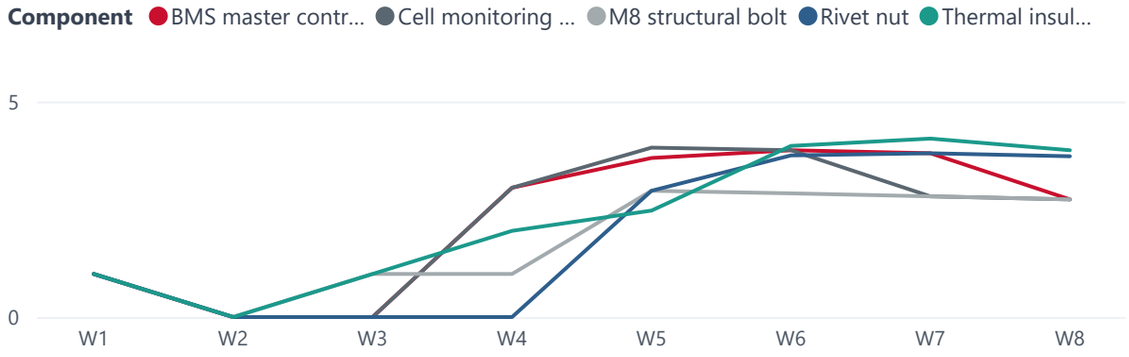
Lost packs, bad case (do nothing)
2,600

Backlog pack-weeks (do nothing)
6,670

Cumulative shortfall vs plan (packs)



Weeks of stock left: top 5 problem parts (do nothing)



How often each part stops the line, by week (do nothing; SR + LR added together)

Component	W2	W3	W4	W5	W6	W7	W8	Total
BMS master controller		11%	13%	3%				**27%**
Cell monitoring board	5%	52%	22%	4%	2%	2%		**87%**
M8 structural bolt		23%	10%	8%	3%	2%	2%	**48%**
Main HV contactor			1%					**1%**
Mica fire barrier sheet			3%					**3%**
Precharge contactor		13%	6%	2%	2%			**23%**
Rivet nut		11%	21%	13%	3%	1%	1%	**50%**
Thermal insulation pad		15%	9%	1%				**25%**
Total	**5%**	**125%**	**85%**	**31%**	**10%**	**5%**	**3%**	**264%**

Simulated data. Placeholder delay parameters.

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Scenario Explorer

Pick policies to compare. Right-click a component in the actions table > Drill through > Component detail.

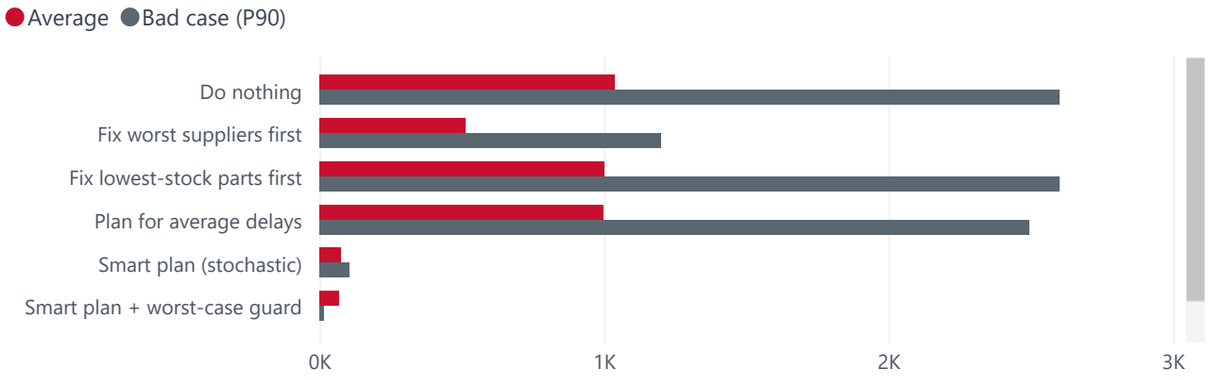
Scenario

Asia ocean-lane port congestion

Strategy (pick any)

- [ ] Do nothing
- [ ] Fix worst suppliers first
- [ ] Fix lowest-stock parts first
- [ ] Plan for average delays
- [ ] Smart plan (stochastic)
- [ ] Smart plan + worst-case guard
- [ ] Perfect foresight (benchmark)

Lost packs by strategy: average vs bad case (worst 10%)



Value of planning for uncertainty

$2M

NOTE

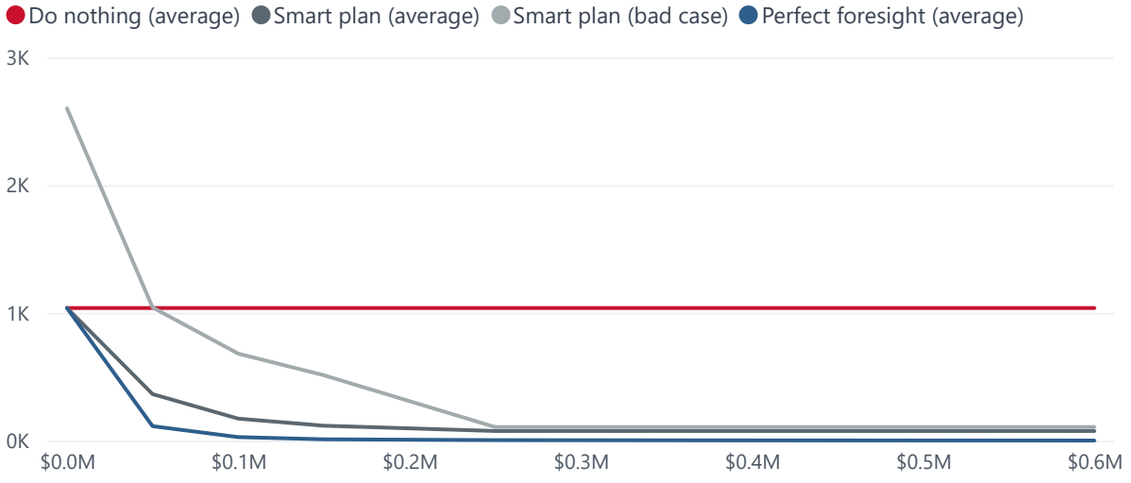
Budget curves exist only for D1 and H4.

For other scenarios the budget chart is empty. It shows lost packs as the mitigation budget grows.

What the plan actually buys

Strategy	Component ID	Component	Action	Week	Quantity	Cos
Fix worst suppliers first	C11	Cell monitoring board	expedite	1	6,000	$40
Fix worst suppliers first	C30	Thermal insulation pad	expedite	1	12,600	$3
Fix worst suppliers first	C31	Mica fire barrier sheet	expedite	1	6,000	$5
Fix worst suppliers first	C32	Aerogel cell spacer	expedite	1	45,000	$20
Fix worst suppliers first	C42	M6 flange bolt	expedite	1	90,000	$1
Fix worst suppliers first	C43	M8 structural bolt	expedite	1	36,000	$1
Fix worst suppliers first	C44	Rivet nut	expedite	1	60,000	$1
Fix lowest-stock parts first	C11	Cell monitoring board	expedite	1	6,000	$40
Fix lowest-stock parts first	C12	BMS master controller	expedite	1	1,500	$36
Smart plan (stochastic)	C11	Cell monitoring board	expedite	1	5,200	$35
Smart plan (stochastic)	C14	Precharge contactor	expedite	1	912	$3
Smart plan (stochastic)	C15	Precharge resistor	expedite	1	5	

Lost packs vs mitigation budget



Simulated data. Placeholder delay parameters.

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Assumptions & Limitations

Everything operational is simulated. Full detail in ASSUMPTIONS.md.

Scenario

Asia ocean-lane port congestion

Static week-0 decisions (no rolling re-planning). Delay parameters are PLACEHOLDERS, not calibrated. Costs and penalties ($2,000 per lost pack, $100 per backlog pack-week, $250,000 budget) are hypothetical.

1. Site, products, plan

- One hypothetical site, weekly buckets, 8-week horizon; SR (LFP) and LR (NMC) packs.
- Plan SR 900 + LR 600 per week (12,000 packs); capacity 1,600/week.
- Lost packs = plan - built at week 8; backlog pack-weeks measure how far and how long production runs behind.

2. Supply

- 20 tier-1 suppliers, 6 tier-2 sources, 4 lanes, 50 components; just-in-time weekly POs.
- Lots can be late, partial, quality-held, or cut by capacity loss / scrap; arrivals after week 8 ignored.

3. Disruptions

- Planner knows which disruption was announced, not its severity or duration.
- D1-D6: truth equals prior. H1-H5: held out, truth deliberately differs from prior.

4. Mitigation actions

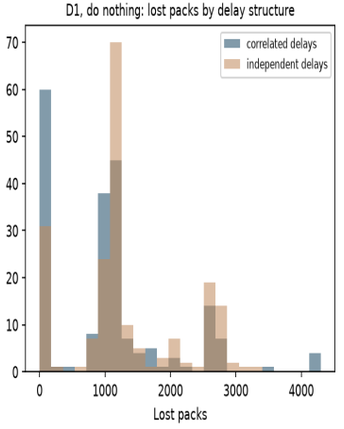
- Expedite (air or team driver), extra inventory, approved alternate supplier.
- Only incremental mitigation cost counts against the $250,000 budget.

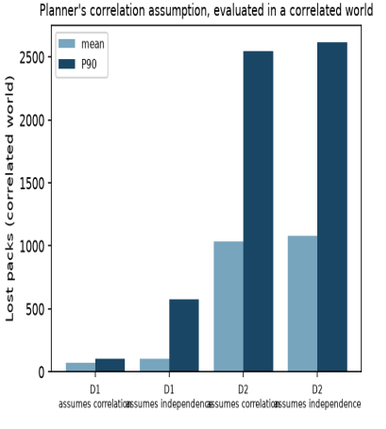
5. Decisions and evaluation

- Static two-stage: all actions decided once at week 0.
- Deterministic LP, Stochastic LP (100 samples), CVaR variant; oracle solves the same LP with perfect knowledge.
- Planning (seed 42) and evaluation (seed 2027) samples are independent.
- Penalty test: at $1K, $2K and $4K per lost pack, the smart plan beats doing nothing in 33 of 33 cases.

6. Known limitations

- Simulated operations; placeholder delays; calibration source is not battery parts.
- No rolling re-planning, no backlog recovery after capacity cuts, no MOQs or integer lots.





Stress tests (all scenarios; not filtered by the scenario slicer)

test	scenario_id	setting	metric	value
Correlation (do nothing)	D1	correlated	mean_lost_packs	1,033.63
Correlation (do nothing)	D1	correlated	p90_lost_packs	2,600.00
Correlation (do nothing)	D1	correlated	p95_lost_packs	2,700.00
Correlation (do nothing)	D1	correlated	prob_any_loss	0.73
Correlation (do nothing)	D1	independent	mean_lost_packs	1,294.56
Correlation (do nothing)	D1	independent	p90_lost_packs	2,600.00
Correlation (do nothing)	D1	independent	p95_lost_packs	2,700.00
Correlation (do nothing)	D1	independent	prob_any_loss	0.86
Correlation (do nothing)	D2	correlated	mean_lost_packs	1,682.93

Correlation-test plans (P5-corr vs P5-indep)

Policy	Action	Quantity	Cost
P5-corr	alternate	64,800	$864
P5-corr	expedite	231,816	$180,382
P5-indep	alternate	28,800	$432
P5-indep	expedite	309,080	$174,573
P5-indep	extra	2,820	$784